SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-012: Dark / Light Theme Toggle

Field	Details
Change Request ID	CR-012
Change Request Title	Dark / Light Theme Toggle
System	Student Management System
Implemented By	Jansan.J
Implementation Status	Completed
Date Implemented	9 October 2026

1. Change Request Summary

1.1 Objective

Implement a global Dark / Light theme toggle across the Student Management System. A theme selected on one page must apply consistently to the Landing Page, Login Page, Admin Dashboard, and Student Dashboard, and the selected preference must persist through navigation and page refresh.

1.2 Existing Behaviour

Before CR-012, the application did not provide a unified theme toggle across the four application areas. Users could not select one global dark or light appearance and have that preference retained consistently when navigating between pages.

1.3 Requested Change

- Provide a reusable theme toggle on the Landing Page, Login Page, Admin Dashboard, and Student Dashboard.
- Share theme state globally using React Context and initialize to light mode when no saved preference exists.
- Persist the selected theme in localStorage under the key app-theme and synchronize the HTML data-theme attribute.
- Apply dark-mode styling through a centralized dark-theme.css stylesheet while preserving existing page functionality.

2. Implementation

2.1 Implementation Summary

CR-012 implements a global theme preference using ThemeContext and ThemeProvider. The provider wraps the application in main.jsx, exposes the current theme and toggleTheme action, and stores the selected value under app-theme in localStorage. The HTML data-theme attribute is synchronized so centralized CSS overrides can update colors across all pages. A reusable ThemeToggle component is placed in the navigation or page header appropriate to each screen. No backend or database changes were required.

2.2 Affected Files

File	Change	Purpose
frontend/src/context/ThemeContext.jsx	New	Provides shared theme state, localStorage persistence, and the document data-theme attribute.
frontend/src/components/ThemeToggle.jsx	New	Reusable button showing Dark Mode / Light Mode and an accessible aria-label.
frontend/src/components/ThemeToggle.css	New	Styles for the reusable theme toggle button.
frontend/src/styles/dark-theme.css	New	Centralized dark-mode CSS overrides (265 lines).
frontend/src/main.jsx	Modified	Wraps App with ThemeProvider.
frontend/src/index.css	Modified	Imports dark-theme.css.
frontend/src/components/Navbar.jsx	Modified	Adds ThemeToggle to Landing Page navigation.
frontend/src/pages/LoginPage.jsx	Modified	Adds ThemeToggle in the top-right corner.
frontend/src/pages/DashboardPage.jsx	Modified	Adds ThemeToggle in Admin Dashboard header actions.
frontend/src/pages/StudentDashboardPage.jsx	Modified	Adds ThemeToggle in Student Dashboard header actions.

2.3 Technical Changes

Global Theme State

- ThemeProvider initializes theme from localStorage.getItem("app-theme"); the default is light when no value is saved.
- toggleTheme switches between light and dark, and the effect writes the active value to localStorage and sets document.documentElement data-theme.
- All application pages use the same React Context state so changing the theme in one place updates the other pages.

Centralized CSS Overrides

The dark-theme.css file uses the [data-theme="dark"] selector to override the appearance of global text and backgrounds, navigation, login elements, landing sections, dashboard headers, cards, forms, tables, profile information, action buttons, and the footer.

Theme Toggle Placement

Page	Placement
Landing Page	Inside Navbar navigation links.
Login Page	Absolute-positioned top-right corner.
Admin Dashboard	Header actions bar beside logout.
Student Dashboard	Header actions bar beside logout.

Dark Mode Colour Palette

Token	Hex Code	Usage
Background	#0f172a	Page backgrounds
Surface	#1e293b	Cards, headers, navbar
Elevated	#334155	Inputs, icons, table headers
Border	#475569	Input borders and dividers
Text Primary	#f8fafc	Headings and strong text
Text Body	#cbd5e1	Paragraph and general text
Text Muted	#94a3b8	Descriptions and labels
Accent	#60a5fa	Links, hover states, labels
Accent BG	#3b82f6	Focus rings

2.4 Implementation Logic

1. The application mounts with the saved app-theme value, or light when there is no saved value.
2. ThemeProvider makes the active theme and toggleTheme action available throughout the application.
3. The user activates the toggle on any supported page.
4. React state changes from light to dark or from dark to light.
5. The updated theme is saved to localStorage and reflected in the HTML data-theme attribute.
6. The centralized CSS rules update the appearance of supported page elements.
7. Navigation to another page or refreshing the page retains the stored theme preference.

2.5 Error and Data Handling

Theme preference is stored in the browser under app-theme. When no preference is found, the application defaults to light mode. Theme state and presentation are handled on the frontend; the source report identifies no backend, API, or database changes for this request.

3. Testing & Verification

3.1 Testing Approach

The source implementation report records manual frontend verification across page switching, theme persistence, and the visibility of interface elements in dark mode.

Testing Types:

- Functional testing
- UI and readability testing
- Navigation and persistence testing
- Regression testing across application pages

Environment: Development

Browser: Not specified in the source report

3.2 Test Cases

Test ID	Test Case	Expected Result	Recorded Result	Status
TC-01	Toggle to dark mode on Landing Page	All pages switch to dark theme.	Global dark theme applied.	Pass
TC-02	Toggle to light mode on Login Page	All pages revert to light theme.	Global light theme restored.	Pass
TC-03	Refresh after selecting dark mode	Dark mode persists using localStorage.	Saved preference persists.	Pass
TC-04	Navigate Landing to Login in dark mode	Login Page renders in dark mode.	Theme carries to Login Page.	Pass
TC-05	Navigate Login to Dashboard in dark mode	Dashboard renders in dark mode.	Theme carries to dashboard.	Pass
TC-06	Toggle on Student Dashboard	Theme applies globally.	Shared theme state updates pages.	Pass
TC-07	Check text in dark mode	Text remains sufficiently readable.	Text readable across tested elements.	Pass
TC-08	Check form inputs in dark mode	Inputs show dark backgrounds and light text.	Inputs remain visible.	Pass
TC-09	Check table rows in dark mode	Table rows remain visible with dark backgrounds.	Table rows visible.	Pass
TC-10	Check icons in dark mode	Icon containers have appropriate dark backgrounds.	Icons remain visible.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	10
Passed	10
Failed	0
Blocked	0
Not Tested	0
Pass Rate	100%

Overall Test Result: Passed

Summary: The source report records all 10 listed manual UI, navigation, persistence, and readability checks as passed.

4. Acceptance Criteria Verification

The criteria below are derived from the functionality and tests documented in the supplied CR-012 implementation report.

ID	Acceptance Criterion	Verification	Status
AC-001	Theme toggle is available on Landing, Login, Admin Dashboard, and Student Dashboard.	Implementation/file review and TC-01, TC-04 to TC-06	Met
AC-002	Initial theme defaults to light when no app-theme preference is stored.	ThemeContext initialization review	Met
AC-003	Users can switch between dark and light mode.	TC-01 and TC-02	Met
AC-004	Selected theme persists after refresh and navigation.	TC-03 to TC-05	Met
AC-005	The HTML data-theme attribute is synchronized with theme state.	ThemeContext implementation review	Met
AC-006	Text, forms, tables, icons, and major page areas remain visible in dark mode.	TC-07 to TC-10	Met
AC-007	Toggle controls provide an accessible aria-label and clear mode text/icon.	ThemeToggle component review	Met
AC-008	Theme functionality is implemented without backend or database changes.	Source file scope review	Met

5. Evidence

Take the screenshots from the running Student Management System and insert them into the labelled evidence areas. Screenshots 1–5 are the core evidence set; Screenshot 6 is recommended to document theme persistence after refresh.

Evidence 01 – Landing Page in Light Mode

Related: AC-001, AC-002, AC-003

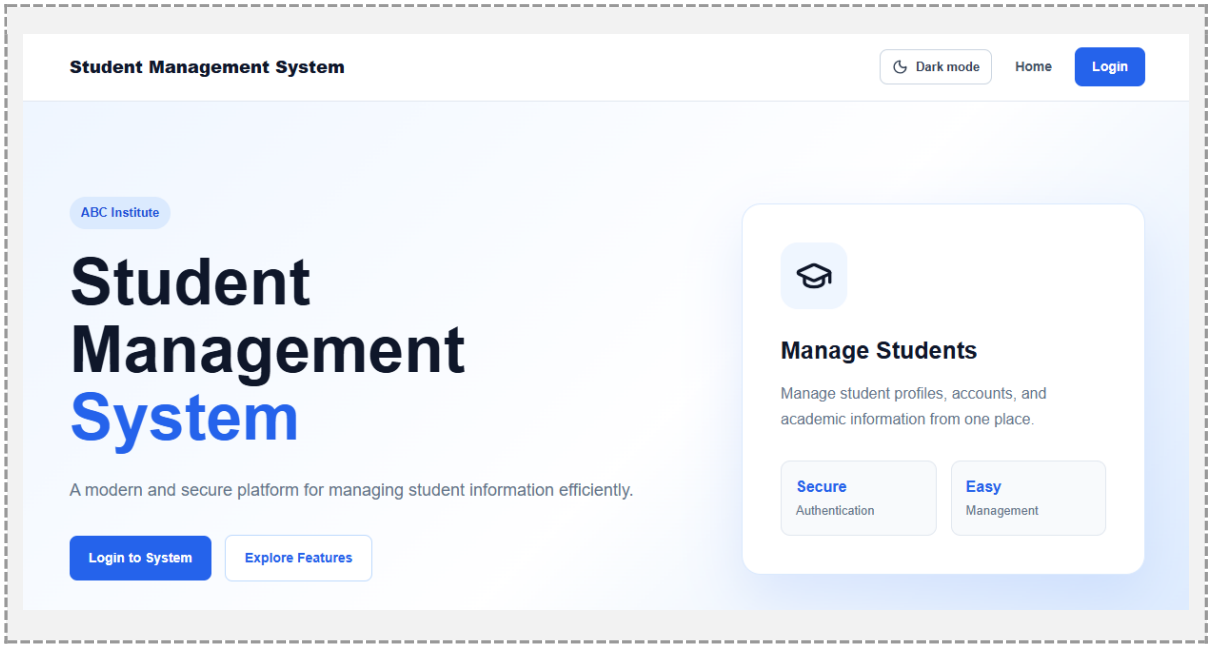


Figure 1: Landing Page in the default light theme with the theme toggle visible and labelled to switch to dark mode.

Evidence 02 – Landing Page in Dark Mode

Related: AC-001, AC-003, AC-006

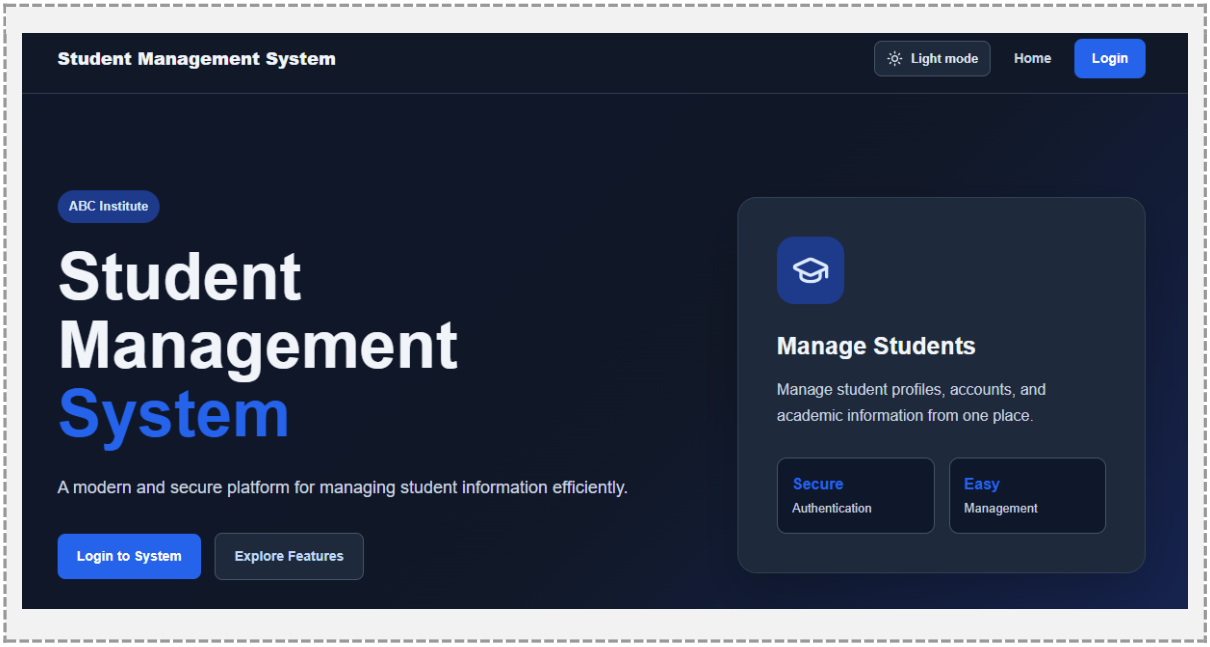


Figure 2: Landing Page after switching to dark mode, showing the toggle label and dark styling across the navigation and landing content.

5. Evidence (continued)

Evidence 03 – Login Page in Dark Mode

Related: AC-001, AC-003, AC-006

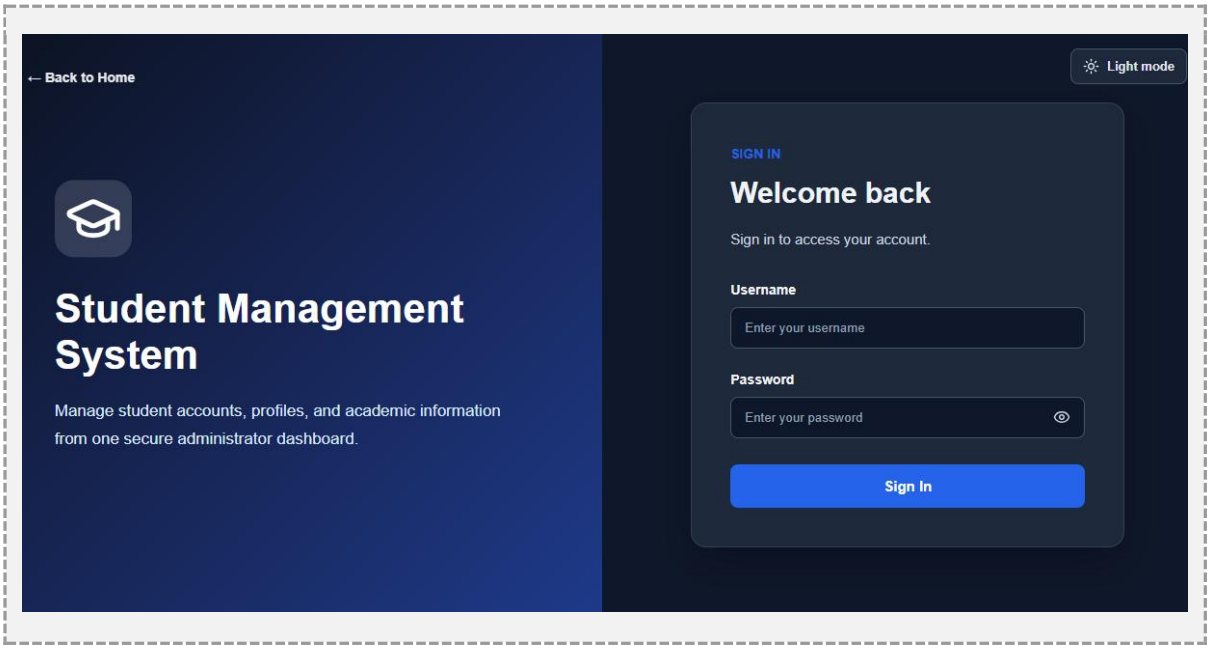


Figure 3: Login Page showing the top-right theme toggle and dark styling of the login panel, form inputs, and buttons.

Evidence 04 – Admin Dashboard in Dark Mode

Related: AC-001, AC-003, AC-006

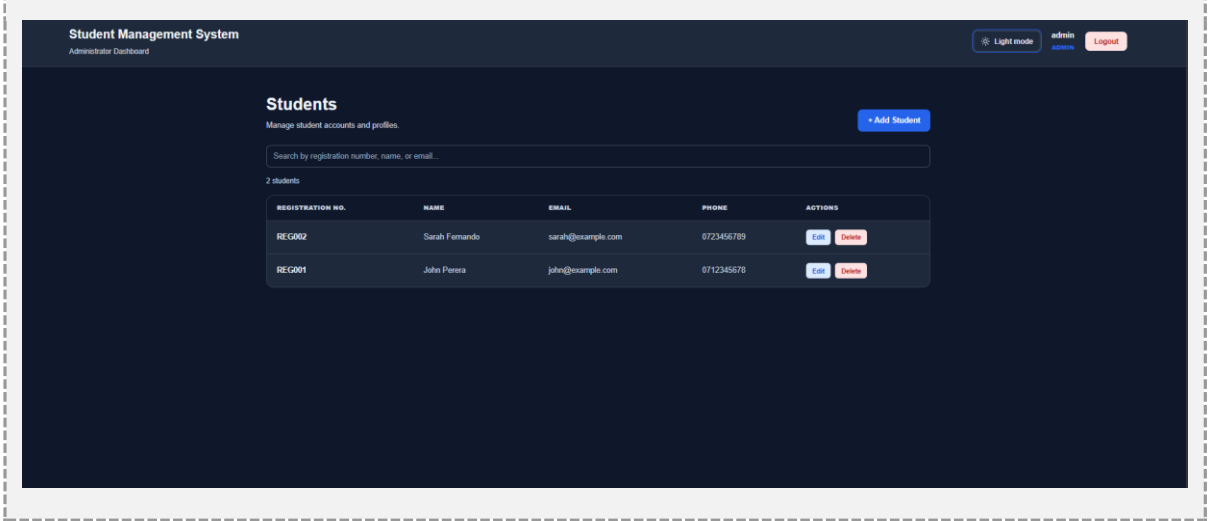


Figure 4: Admin Dashboard showing the theme toggle beside logout and dark styling on the student table, search field, and controls.

5. Evidence (continued)

Evidence 05 – Student Dashboard in Dark Mode

Related: AC-001, AC-003, AC-006

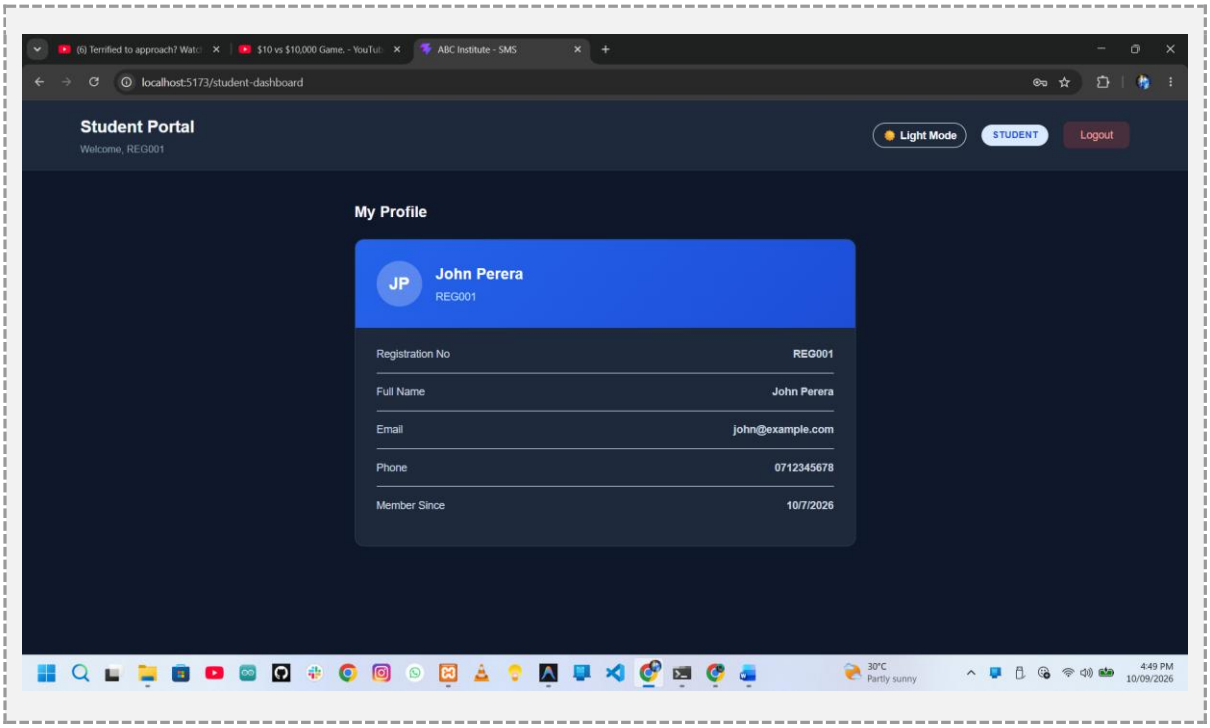


Figure 5: Student Dashboard showing the theme toggle in the header and dark styling on profile details and the logout button.

Evidence 06 – Theme Persistence After Refresh

Related: AC-004

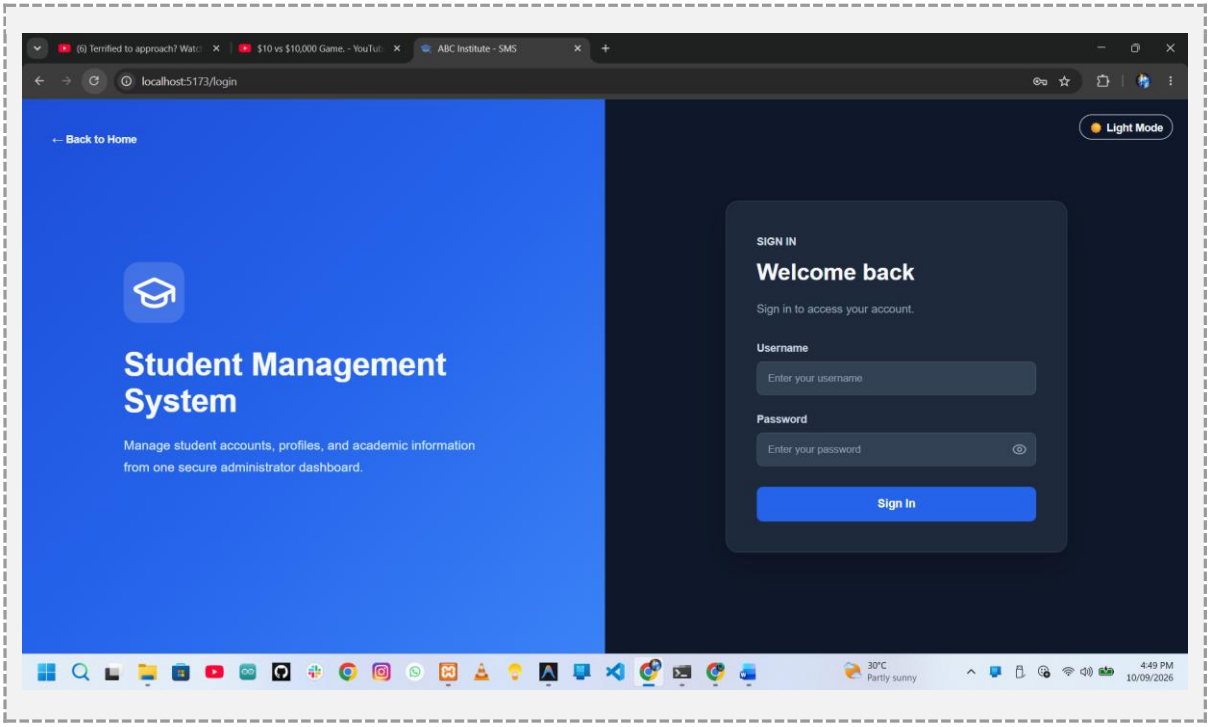


Figure 6: A supported page after refresh showing that the previously selected dark theme remains active.

6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed (10/10 documented checks)
Acceptance Criteria	All Met
Backend / Database Changes	None required
Final Status	Completed

6.1 Final Summary

CR-012 implements a global Dark / Light theme toggle across the Landing Page, Login Page, Admin Dashboard, and Student Dashboard. React Context provides shared state, localStorage persists the selected theme under app-theme, and the HTML data-theme attribute enables centralized CSS overrides. The source report records ten successful verification checks and identifies no backend or database modifications.

6.2 Issues / Limitations

No significant implementation issues are identified in the supplied Markdown report. Insert the actual screenshots into the evidence placeholders before final submission.

Implemented By: Developer

Date: 9 October 2026

Signature: _____